WORKSHEET · SCRATCH (GR 6–8) BEGINNER: CODE FOUNDATIONS

Lesson 4: Loops and Polygons

Module 2 | 45 minutes | Counted loops (iteration) | Grades 6–8 · ages 11–14

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. iteration ___ A. Make one solution work for many cases.
b. counted loop ___ B. How far you turn at each corner.
c. exterior angle ___ C. A loop that repeats a fixed number of times.
d. generalise ___ D. Repeating a set of instructions.

2. Plan your code

Before you build it: Hexagon with a loop: draw a closed regular hexagon (6 sides) using a repeat loop. Each side is 60 steps. Use no more than 10 blocks. Write or draw the blocks you will use.

3. Pentagon turn

What turn angle draws a regular pentagon (5 sides)? Circle one.

a. A. 72°
b. B. 108°
c. C. 60°

4. Plan your code

Before you build it: Polygon machine: set a variable sides to 8. Use repeat (sides) with move 40 and turn right (360 / sides) to draw a closed octagon. Then you could change sides to draw any polygon. Write or draw the blocks you will use.

5. Loops in Python

In Python, what does this print? Code: for i in range(3): ↵ print("hip") ↵ print("hooray")

a. A. hip hip hip hooray
b. B. hip hooray hip hooray hip hooray
c. C. hip hooray

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can use a counted loop to remove repetition.
☐ I can calculate the turn for any polygon.
☐ I can use a variable and operator to generalise.